

Quantum Teleportation: Complete Mathematical Derivation and Quantum Circuit

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1 Introduction

Quantum teleportation is a quantum communication protocol through which an unknown quantum state can be transferred from one location to another using

- quantum entanglement,
- Bell-state measurement,
- and classical communication.

The protocol was proposed by Bennett *et al.* in 1993.

Unlike classical communication, the actual particle is not transmitted. Only the quantum information encoded in the state is transferred.

2 Representation of a Qubit

A general single-qubit pure state is represented by

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad (1)$$

where

$$\alpha, \beta \in \mathbb{C} \quad (2)$$

and normalization requires

$$|\alpha|^2 + |\beta|^2 = 1 \quad (3)$$

In vector form,

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \quad (4)$$

3 Bell States

The Bell states form an orthonormal basis for two-qubit systems.

3.1 Bell Basis

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \quad (5)$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle) \quad (6)$$

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle) \quad (7)$$

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle) \quad (8)$$

4 Entangled Pair Shared Between Alice and Bob

Suppose Alice and Bob share the Bell state

$$|\Phi^+\rangle_{23} = \frac{1}{\sqrt{2}} (|00\rangle_{23} + |11\rangle_{23}) \quad (9)$$

where

- qubit 2 belongs to Alice,
- qubit 3 belongs to Bob.

Alice additionally possesses an unknown state

$$|\psi\rangle_1 = \alpha|0\rangle_1 + \beta|1\rangle_1 \quad (10)$$

5 Initial Three-Qubit State

The total system is

$$|\Psi\rangle_{123} = |\psi\rangle_1 \otimes |\Phi^+\rangle_{23} \quad (11)$$

Substituting,

$$|\Psi\rangle_{123} = (\alpha|0\rangle_1 + \beta|1\rangle_1) \otimes \frac{1}{\sqrt{2}} (|00\rangle_{23} + |11\rangle_{23}) \quad (12)$$

$$= \frac{1}{\sqrt{2}} (\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle) \quad (13)$$

6 Rewriting in Bell Basis

Now express the first two qubits in Bell basis.

Using

$$|00\rangle = \frac{1}{\sqrt{2}} (|\Phi^+\rangle + |\Phi^-\rangle) \quad (14)$$

$$|11\rangle = \frac{1}{\sqrt{2}} (|\Phi^+\rangle - |\Phi^-\rangle) \quad (15)$$

$$|01\rangle = \frac{1}{\sqrt{2}} (|\Psi^+\rangle + |\Psi^-\rangle) \quad (16)$$

$$|10\rangle = \frac{1}{\sqrt{2}} (|\Psi^+\rangle - |\Psi^-\rangle) \quad (17)$$

we obtain

$$|\Psi\rangle_{123} = \frac{1}{2} \left[|\Phi^+\rangle_{12}(\alpha|0\rangle + \beta|1\rangle)_3 \right. \quad (18)$$

$$+ |\Phi^-\rangle_{12}(\alpha|0\rangle - \beta|1\rangle)_3 \quad (19)$$

$$+ |\Psi^+\rangle_{12}(\beta|0\rangle + \alpha|1\rangle)_3 \quad (20)$$

$$\left. + |\Psi^-\rangle_{12}(\beta|0\rangle - \alpha|1\rangle)_3 \right] \quad (21)$$

This is the central equation of quantum teleportation.

7 Bell-State Measurement

Alice performs Bell-state measurement on qubits 1 and 2.

The system collapses into one of the four Bell states.

Depending on Alice's measurement outcome, Bob's qubit becomes:

Alice's Measurement	Bob's State
$ \Phi^+\rangle$	$\alpha 0\rangle + \beta 1\rangle$
$ \Phi^-\rangle$	$\alpha 0\rangle - \beta 1\rangle$
$ \Psi^+\rangle$	$\beta 0\rangle + \alpha 1\rangle$
$ \Psi^-\rangle$	$\beta 0\rangle - \alpha 1\rangle$

8 Classical Communication

Alice communicates two classical bits to Bob corresponding to her Bell-state measurement outcome.

Bob then applies an appropriate unitary operator.

9 Pauli Operators

9.1 Identity Operator

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (22)$$

9.2 Pauli-X Operator

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (23)$$

Action:

$$X|0\rangle = |1\rangle \quad (24)$$

$$X|1\rangle = |0\rangle \quad (25)$$

9.3 Pauli-Z Operator

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (26)$$

Action:

$$Z|0\rangle = |0\rangle \quad (27)$$

$$Z|1\rangle = -|1\rangle \quad (28)$$

10 Bob's Recovery Operations

Measurement Outcome	Bob's Operation
$ \Phi^+\rangle$	I
$ \Phi^-\rangle$	Z
$ \Psi^+\rangle$	X
$ \Psi^-\rangle$	XZ

11 Verification of Recovery

11.1 Case 1

If Bob receives

$$\alpha|0\rangle + \beta|1\rangle \quad (29)$$

he applies

$$I \quad (30)$$

and obtains

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad (31)$$

11.2 Case 2

If Bob receives

$$\alpha|0\rangle - \beta|1\rangle \quad (32)$$

then

$$Z(\alpha|0\rangle - \beta|1\rangle) = \alpha Z|0\rangle - \beta Z|1\rangle \quad (33)$$

$$= \alpha|0\rangle + \beta|1\rangle \quad (34)$$

11.3 Case 3

If Bob receives

$$\beta|0\rangle + \alpha|1\rangle \quad (35)$$

then

$$X(\beta|0\rangle + \alpha|1\rangle) = \beta X|0\rangle + \alpha X|1\rangle \quad (36)$$

$$= \beta|1\rangle + \alpha|0\rangle \quad (37)$$

$$= \alpha|0\rangle + \beta|1\rangle \quad (38)$$

11.4 Case 4

If Bob receives

$$\beta|0\rangle - \alpha|1\rangle \quad (39)$$

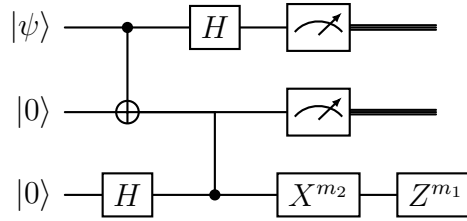
then

$$XZ(\beta|0\rangle - \alpha|1\rangle) = X(\beta|0\rangle + \alpha|1\rangle) \quad (40)$$

$$= \alpha|0\rangle + \beta|1\rangle \quad (41)$$

12 Quantum Circuit of Teleportation

12.1 Teleportation Circuit



13 Circuit Explanation

13.1 Step 1: Entanglement Generation

Initially,

$$|00\rangle \quad (42)$$

Applying Hadamard on qubit 2,

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad (43)$$

Thus,

$$\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \quad (44)$$

Applying CNOT,

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\Phi^+\rangle \quad (45)$$

13.2 Step 2: Bell Measurement

Alice applies

- CNOT gate,
- Hadamard gate,
- measurement.

This converts Bell basis into computational basis.

13.3 Step 3: Classical Communication

Alice sends bits

$$m_1, m_2 \quad (46)$$

to Bob.

13.4 Step 4: Conditional Operations

Bob applies

- X correction depending on m_2
- Z correction depending on m_1

14 Matrix Representation of Gates

14.1 Hadamard Gate

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (47)$$

14.2 CNOT Gate

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad (48)$$

15 No-Cloning Theorem

Quantum teleportation does not violate the no-cloning theorem.

Suppose there exists a unitary operator U such that

$$U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle \quad (49)$$

for all states.

Consider two states $|\psi\rangle$ and $|\phi\rangle$.

Then

$$U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle \quad (50)$$

$$U|\phi\rangle|0\rangle = |\phi\rangle|\phi\rangle \quad (51)$$

Taking inner products,

$$\langle\psi|\phi\rangle = (\langle\psi|\phi\rangle)^2 \quad (52)$$

which is impossible for arbitrary states.

Hence universal cloning cannot exist.

16 Advantages of Quantum Teleportation

- Secure quantum communication
- Essential for quantum networks
- Basis of quantum repeaters
- Enables distributed quantum computation
- Transfers unknown quantum states

17 Limitations

- Requires entanglement resource
- Sensitive to decoherence
- Requires classical communication
- Experimentally challenging

18 Applications

- Quantum internet
- Quantum repeaters
- Quantum cryptography

- Satellite quantum communication
- Distributed quantum computing

19 Conclusion

Quantum teleportation is one of the most important protocols in quantum information science. The protocol transfers an unknown quantum state using shared entanglement and classical communication.

Mathematically, teleportation demonstrates the power of Bell-state decomposition, quantum measurement, and unitary correction operations.

The protocol forms the basis for future quantum communication networks and large-scale quantum technologies.

20 References

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